Appendix A: Data appendix

A.1 District head appointments

I identify 586 individuals for a total of 751 unique district head tenures (678/531 end up in the panel) using a mix of online sources, including gov.kz biography pages for bureaucrats who are still serving, Zakon's 'Paragraf' system (https://online.zakon.kz/) and Daniel Ashimbayev's *Kto est' kto* (https://centrasia.org/person.php), and local news reports. This sample excludes periods where heads are appointed on a temporary basis before a new head is appointed as well as those heads directly elected on a trial basis from November 2023. At the time of data collection in early 2026, 152 of the 678 panel terms were still in office, that is, the head had not departed since the end of 2024.

I also identify the tenures of the 61 regional governors who appointed each district head from Wikipedia and create a variable for whether the governor who appointed the district head is still in office in that quarter.[A1]

A.2 Senior bureaucrats data

I identify the appointments of the most senior district bureaucrats from 2018 onwards. These appointments are not publicly reported by state institutions or the media. Instead, I rely on computational processing of signatures from public procurement documents, which state institutions must digitally publish for most purchases

[A1] See 'Spisok akimov oblastey Kazakhstana [*List of akims of the regions of Kazakhstan*]', Wikipedia, https://ru.wikipedia.org/wiki/Список_акимов_областей_Казахстана.

and which are signed by a senior bureaucrat.[A2] I code new appointments when the name of that bureaucrat changes between documents.

A.2.1 Identifying district institutions

I identify government institutions including district heads' offices, district departments, and subdistrict apparatuses (excluding Astana, Almaty, and Shymkent). I identify these institutions from a database of legal entities in Kazakhstan, which includes all business and government institutions registered with tax authorities. I create a dataset of all legal entities by downloading the Excel files of currently registered (as of January 2025) legal entities by each region.

From this dataset, I include institutions identified in two ways:

1. All entities with the classification code (OKED, General Classifier of Economic Activities) '84113' (local government, which covers all *rayon*-level bodies) or '84114' (rural and town government, which covers sub-district bodies).[A3]

2. First, I subset the remaining entities to those with the ownership classification (KFC, Classifier of Forms and Types of Ownership) '13' (Communal ownership, which covers all sub-republican government structures).[A4] Second, I use regular expressions (regex) to identify those with patterns in their names that I have identified as being used by these institutions. In particular, they start with the phrase '[communal] state institution' (in Russian) or its abbreviation ([K]GU) and include the terms 'Department', or 'Rayon department' or 'City department' directly following that. They do not contain the term 'police'.

[A2] See the procurement website at https://goszakup.gov.kz/ru/.
[A3] See https://stat.gov.kz/ru/classifiers/statistical/21/.
[A4] See https://stat.gov.kz/ru/classifiers/statistical/22/.

Finally, I clean these database by removing republican- and regional-level institutions that either were given the incorrect OKED code or were caught inadvertently by the regex method. This involves removing specific institutions which I have manually identified, as well as those matching a second set of regex rules. For example, these rules include removing names including the phrase 'Ministry of Finance' (which indicates tax collection bodies), 'territorial inspectorate', and 'regional sub-commission'. I additionally remove locally elected representative bodies.

There are three reasons that I likely do not have the exact universe of institutions. First, since 4,818 institutions result from this process, I do not manually verify that all which remain are the top-level district departments which I intend to study (rather than being sub-departments or branches of regional government). However, an overview of the data suggests that this process has captured only those institutions. Second, some institutions stopped operating within the period I study. While it would be possible to include at least some of these using historical records, here I show results only using currently operating institutions for simplicity and to avoid a change in the number of institutions over time. I estimate that this would only increase the number of active institutions by about 4.5%. Third, my process does not capture those institutions which were incorrectly not assigned the OKED 84113 or 84114 and which also were wrongly not recorded with the KFC 13 or have a KFC of 13 but do not match the regex expression I define (due to a misspelling, a few of which I do manually identify). However, these would require a number of mistakes by the Kazakhstani state as well as escaping detection in my own manual checks.

Third, almost every institution in this list has available procurement data. I observe at least one signatory for 4,575 of them, and appointments in 2,903, 2,827, and 2,589 institutions under the 90-, 180-, and 365-day rules respectively. This gap is for three

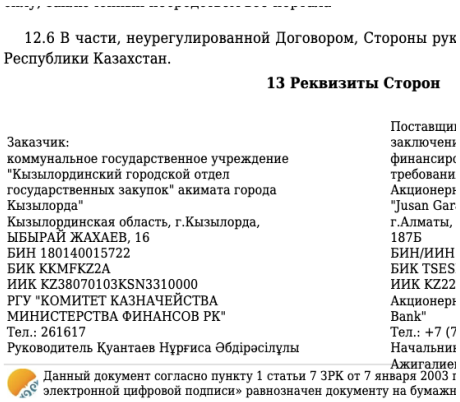
reasons. First, as below, I discard the first observed signatory in each institution. Second, some institutions (especially subdistricts) have very turnover in

I am only able to identify appointments in 2,589 institutions which have signed documents. This is a core limitation of the method, which relies on institutions frequently posting documents and therefore is more likely to exclude institutions which rarely engage in public procurement (e.g., especially those with responsibility for cultural issues). On the other hand, it does mean my data contain the most 'active'—so in some sense, the most important—district institutions.

A.2.2 Identifying bureaucrats

I identify public procurement documents signed by each of these institutions since 2016 by scraping the government e-procurement portal https://www.goszakup.gov.kz/. The signatory on these documents is the name of a senior bureaucrat. Who that is depends on whether the contracting institution is the district government as a whole, one of the executive bodies financed from the local budget, or the office of a subdistrict head. Contracts with the whole district government are signed by the head of the district head's office, those with executive bodies are signed by the head of the executive body, and those with subdistrict heads are signed by the district head themselves.[A5]

[A5] Unfortunately, deputy district heads—who explicitly act as the head's trusted lieutenants within districts—do not systematically sign documents and so I cannot collect their names.



12.6 В части, неурегулированной Договором, Стороны рук
Республики Казахстан.

13 Реквизиты Сторон

Заказчик:	Поставщи
коммунальное государственное учреждение	заключен
"Кызылординский городской отдел	финансир
государственных закупок" акимата города	требовани
Кызылорда"	Акционер
Кызылординская область, г.Кызылорда,	"Jusan Gar
ЫБЫРАЙ ЖАХАЕВ, 16	г.Алматы,
БИН 180140015722	187Б
БИК KKMFKZ2A	БИН/ИИН
ИИК KZ38070103KSN3310000	БИК TSES
РГУ "КОМИТЕТ КАЗНАЧЕЙСТВА	ИИК KZ22
МИНИСТЕРСТВА ФИНАНСОВ РК"	Акционер
Тел.: 261617	Bank"
Руководитель Қуантаев Нұрғиса Әбдірәсілұлы	Тел.: +7 (7
	Начальни
	Ажигалие

Данный документ согласно пункту 1 статьи 7 ЗРК от 7 января 2003 г
электронной цифровой подписи» равнозначен документу на бумажн

Figure A1: Example 'signature section' on signed public procurement contract

First, I download the universe of public procurement contract PDFs. These are final agreements for all procurement projects, regardless of whether they are competitively awarded. I download signed agreements in PDF form for each contract. Each document contains a signature section (see the example in Figure A1) containing the name of a signatory from the customer (i.e., the government institution) and that of the supplier. I use PDF extraction tools and regex to identify the customer's signatory's name by (a) isolating the page containing this section, (b) slicing it in half to keep the customer's signature block, and (c) selecting the last line of the block, extracting from it both the title and the name. Second, I use the title and name recorded on pages advertising lots for tender (for non-single supplier purchases). In this case, I can simply scrape the name and title from the HTML.

I consider the title of the person signing the document, discarding the document if it is not one of 'head' or (for subdistricts) 'akim'. I deliberately drop documents signed by 'acting' or deputy heads. This process results in a list of hundreds or thousands of name-date pairs for each institution. I compress these into a name-first date format. First, I identify names that match exactly or have the same surname and first initial

(sometimes, signatories are recorded in this format). Second, I minimise the problems which might arise from problems with my name extraction method by filtering out obviously bogus names—those that include phrases like 'electronic', 'city', 'head', or 'document'. Third, I record a 'spell' for the first time each name appears in an unbroken sequence. This leads to an initial collection of approximately 25,400 bureaucrat spells since 2016. First, my research and checking the names of a number of signatories confirm that this name typically is—as the title part of the signatory block suggests—the relevant head of the institution. Occasionally, however, documents appear to be signed by someone covering for that person, for example. Second, while the PDFs are typically in a very similar form, there are sometimes small variations or mistakes in how each signatory's name is presented that result in me extracting an incorrect name. In both cases, as I detail below, I attempt to mitigate by removing names which only appear in a few documents.

Note that sometimes a bureaucrat's last few documents are posted weeks after the next bureaucrat's first contract. I interpret that as a sign that the outgoing bureaucrat has signed them in advance as they tidy up their procurement tasks before departing office, rather than as indicating the next bureaucrats' first contract date (which I use as their appointment date) is incorrect.

Next, I extensively filter this set of names so as to include only the spells which are most likely to be the permanently appointed senior district bureaucrats who I am interested in. First, for all institutions, I remove the first observed bureaucrat. This ensures that I am coding new appointments rather than the initial application of procurement transparency rules. I use only people appointed from 2018 onwards (about 22,200). By this point, institutions were consistently using the public procurement system. Second, I attempt to deal with the problem of documents being

signed by short-term stand-ins for the main bureaucrat, some of whom remain even after filtering out those with an 'acting' title.[A6] I remove all spells contained within another. The typical manifestation of this is that a long-term office holder's name is replaced by another for the period of a few weeks, before the long-term office holder returns. I interpret this as being when the long-term holder is on leave or otherwise temporarily unavailable to approve documents. This includes approximately 4,960 (90 days), 4,830 (180), and 4,560 (365) observations. I remove all spells of under 90, 180, or 365 days. This step removes those in office for unrealistically short periods of time or which I interpret as temporary appointments as acting cover (perhaps before the permanent appointment is made). At the same time, I apply limits to the number of contracts signed: a bureaucrat must appear on 10 (90 days), 20 (180 days), or 30 (365 days) contracts to be counted. Note that when limiting the number of contracts signed, I use the count of PDFs signed only, since the advert HTML is much more available after 2020 (due to revised competition rules) and so bureaucrats would be much more likely to be recorded after this date. As such, the advert signatures are used to improve appointment dates only. Third, I correct about 1,290 dates where the procurement date and the date I can extract from gov.kz biographies do not match (see Section A.2.3 below). Finally, I remove between 190 and 780 appointments for which I do not have the tenure recorded of the appointing district head.

In total, 5,617 (90 days), 4,796 (180), and 3,628 (365) appointments remain after this cleaning process. From these data, I create a district-quarter panel. The choice of the quarter was for two reasons. First, it helps mitigate the concerns that, especially in institutions which sign relatively few documents, large gaps between signing mean I

[A6] I suspect that, in many cases, acting heads and other non-head employees do not change the title in the procurement system to match their position.

am more confident about the accuracy at this level. Second, it allows me to match the frequency of these data with the quarterly all-employees data provided by the national statistics agency. To panel the data, I create a count of the number of bureaucrats with their first recorded date in each district-quarter. I link this with the district head data, adding to the main dataset a count of the quarterly number of appointments each head makes. Additionally, I include the number of institutions operating in a district, calculated by determining whether the quarter is within the interval between the first and last contract posted by each institution.

One concern here is that the appointment date actually depends on when each bureaucrat signs their first contract. If that is some time after they take office, it may be inaccurate. A related concern for my inference is that my measure of appointments might be driven by a new head taking office and simply leading their subordinates to sign more documents, with a number of previously appointed but inactive bureaucrats then appearing in the data. One factor assuaging this concern is that there are no documents for only about 2.8% of institution-quarters, and only 362 of the 3,628 appointments under the 365-day rule occur in the quarter immediately following such an empty quarter (so the level of inactivity is low).

A.2.3 Verification

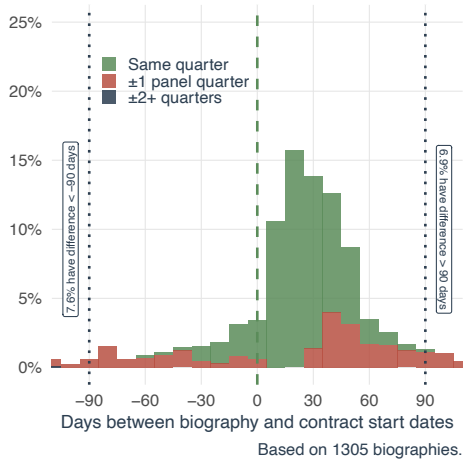


Figure A2: Comparison between procurement dates and dates extracted from available biographies with `GPT-5.4`

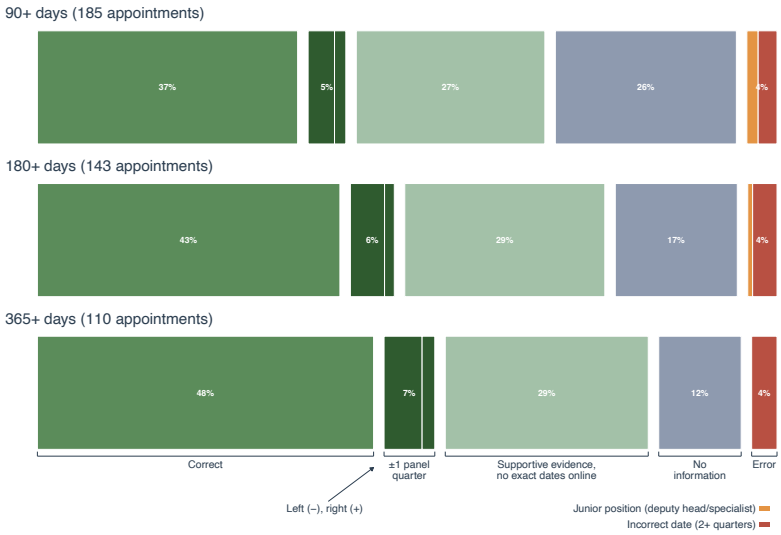


Figure A3: Hand verification of positions and start dates for a sample of 200 appointments (after correction with biography dates)

First, I draw on a corpus of official biographies of the current holders of these roles (published on gov.kz). I can match, by name and position, 2,169 appointments

under the 365-day rule to detailed biographies. I then use an LLM to extract, at the monthly resolution, their official start dates; for 1,305 appointment–biography pairs the biography yields a usable start date to compare.[A7] Figure A2 shows that the procurement dates are mostly in the same quarter as the 'true' dates. Where there are differences, I use the LLM dates to correct the procurement dates. Second, I hand-check a sample of 200 appointments using all available online information. In many cases, it is impossible to verify exactly when someone held a position or even find evidence that they were a civil servant at all. I interpret this as evidence of the value of this sort of data collection, on what are otherwise low-profile and so hard to observe positions. Where dates *are* available, Figure A3 again shows that the procurement dates are reliable, signatures capture department or subdistrict heads, and that almost all of what error remains is small (appointment dates are within a quarter).

After these verification exercises, I believe the data capture (albeit imperfectly) the dependent variable of interest—that is, appointment dates of senior district bureaucrats. Such large-scale data on this case are, as far as I am aware, unavailable from any other source.

A.3 Payroll data

I use data on hiring, separation, and overall headcount in district and subdistrict governments from the national statistics agency (subdistrict statistics are excluded from the non-subdistrict sample).[A8] These data are taken from quarterly reports

[A7] These often capture 'full' appointment rather than when bureaucrats started service as acting officials, which may account for some of the imprecision.

[A8] The data are available from the agency's data portal *Taldau*, see https://taldau.stat.gov.kz/ru/, with indicators 252301, 252302, 251401.

made by the district institutions themselves. The metadata presented alongside the measures claim that they are based on household labour surveys (Form T-001). However, the presented data are (a) implausibly precise for a household labour survey, (b) include information that could not be derived from household surveys, such as average employment counts, but match data collected by *employer* surveys (Form T-1), and (c) are elsewhere described in terms matching Form T-1.

These data do not provide information on the seniority of new employees or which part of district government they work in. For most district-quarters, the source supplies raw counts. In months with very few hires or separations, I recover values which are suppressed for confidentiality (reported as 'x') using the less precise rate measure.Additionally, the hiring and separation measures never contain explicit zeros but instead report missing values ('-'). I infer that these missing values actually mean that no hiring or firing took place if values are missing for both these measures *and* the headcount.[A9] The hiring and separation series remain missing in about 7.5% and 7.6% of district-quarters respectively; in about 3.0% all three series are missing after these exercises.

[A9] The assumption here is that, given they are derived from the same survey, all three should either be present or missing at the same time.

Appendix B: Supplementary Material

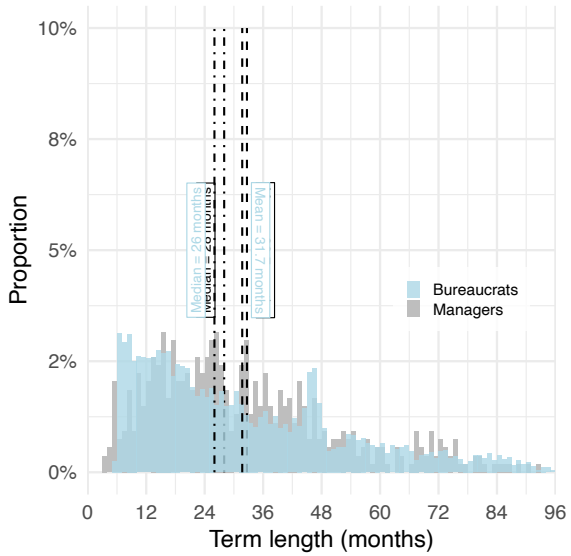
B.1 Case and Data

B.1.1 Case context

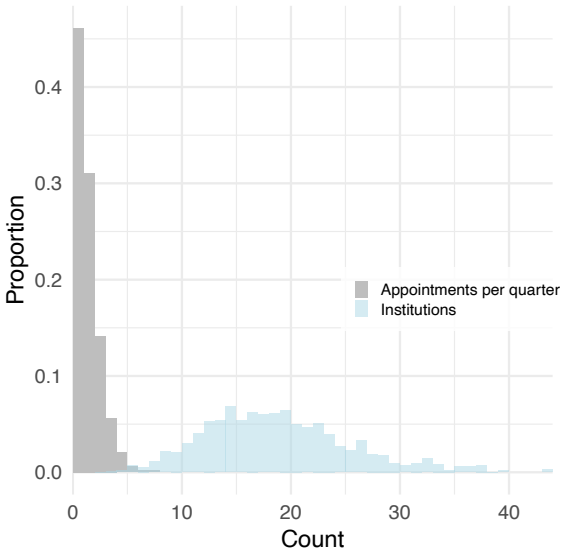
V-Dem data

Figure 1 plots V-Dem's measures of personalism in access to bureaucratic office against restrictions on state employment in electoral autocracies.[B1]. The variables are the country-year means for the period between 2018 and 2023 (the latest V-Dem observation), with the exception of the two Kazakhstan datapoints for the start and end of this period. The trend is the linear trend with a 95% confidence band. Specifically, the plot uses:

- For 'Partisanship in appointments': *Access to state jobs by political group* rescaled to original scale (`v2peasjpol_osp`) (Coppedge et al. 2025, 228–229).
- For 'Criteria for appointment': *Criteria for appointment decisions in the state administration* rescaled to original scale (`v2stcritrecadm_osp`) (200–201).

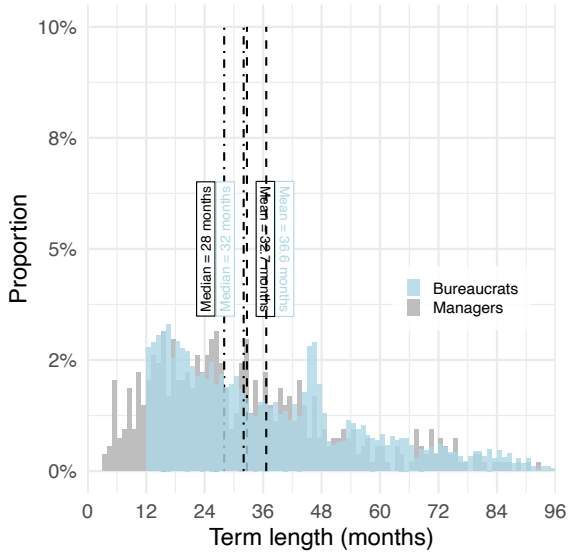


(a) The tenure of district heads and of bureaucrats

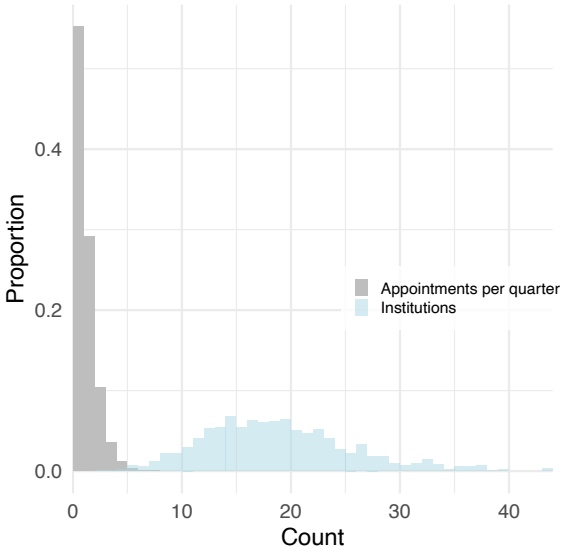


(b) Appointments and institutions

Figure B1: Heads, bureaucrats, and appointments (180-day subset)



(a) The tenure of district heads and of bureaucrats



(b) Appointments and institutions

Figure B2: Heads, bureaucrats, and appointments (365-day subset)

B.1.2 Descriptives

Table B1: Descriptive statistics for baseline models, by measurement rule

Variable	Frequency	Rule	N	Mean	SD	Min	Max	Median
Year	Annual	90 days	4994	NA	NA	2018	2024	2021
		180 days	4994	NA	NA	2018	2024	2021
		365 days	4994	NA	NA	2018	2024	2021
		365/no subdistrict	5137	NA	NA	2018	2024	2021
Senior bureaucrat appts	Quarterly	90 days	4994	1.04	1.19	0	8	1
		180 days	4994	0.89	1.08	0	7	1
		365 days	4994	0.68	0.94	0	7	0
		365/no subdistrict	5137	0.43	0.71	0	7	0
Appts per position	Quarterly	90 days	4994	0.06	0.07	0	0	0
		180 days	4994	0.05	0.06	0	0	0
		365 days	4994	0.04	0.05	0	0	0
		365/no subdistrict	5137	0.04	0.07	0	1	0
N senior bureaucrat positions	Quarterly	90 days	4994	18.28	6.55	3	44	18
		180 days	4994	18.28	6.55	3	44	18
		365 days	4994	18.28	6.55	3	44	18
		365/no subdistrict	5137	9.81	3.18	1	38	9
Local government payroll	Quarterly	90 days	4994	270.04	124.36	51	1214	244
		180 days	4994	270.04	124.36	51	1214	244
		365 days	4994	270.04	124.36	51	1214	244
		365/no subdistrict	5137	140.97	185.66	17	2283	95
Quarters since akim took office	Quarterly	90 days	4994	8.79	6.29	1	39	7
		180 days	4994	8.79	6.29	1	39	7
		365 days	4994	8.79	6.29	1	39	7
		365/no subdistrict	5137	8.76	6.27	1	39	7
Log population	Annual	90 days	4994	10.69	0.92	9	13	11
		180 days	4994	10.69	0.92	9	13	11
		365 days	4994	10.69	0.92	9	13	11
		365/no subdistrict	5137	10.73	0.95	9	13	11

[B1]That is, countries which were electoral autocracies for the whole period according to the Regimes of the World variable in V-Dem (Lührmann, Tannenberg, and Lindberg 2018)

Table B2: Descriptive statistics for all-employees models, by measurement rule

Variable	Frequency	Rule	N	Mean	SD	Min	Max	Median
Year	Annual	90 days	4789	NA	NA	2018	2024	2021
		180 days	4789	NA	NA	2018	2024	2021
		365 days	4789	NA	NA	2018	2024	2021
		365/no subdistrict	4906	NA	NA	2018	2024	2021
Senior bureaucrat appt rate	Quarterly	90 days	4789	0.06	0.07	0	0	0
		180 days	4789	0.05	0.06	0	0	0
		365 days	4789	0.04	0.05	0	0	0
		365/no subdistrict	4906	0.04	0.07	0	1	0
Payroll hire rate	Quarterly	90 days	4789	0.05	0.04	0	1	0
		180 days	4789	0.05	0.04	0	1	0
		365 days	4789	0.05	0.04	0	1	0
		365/no subdistrict	4906	0.05	0.05	0	1	0
Payroll hires	Quarterly	90 days	4789	12.42	13.14	0	242	9
		180 days	4789	12.42	13.14	0	242	9
		365 days	4789	12.42	13.14	0	242	9
		365/no subdistrict	4906	7.56	20.54	0	1035	4
Payroll dismissal rate	Quarterly	90 days	4783	0.05	0.04	0	1	0
		180 days	4783	0.05	0.04	0	1	0
		365 days	4783	0.05	0.04	0	1	0
		365/no subdistrict	4902	0.05	0.05	0	1	0
Payroll dismissals	Quarterly	90 days	4783	12.11	13.56	0	495	9
		180 days	4783	12.11	13.56	0	495	9
		365 days	4783	12.11	13.56	0	495	9
		365/no subdistrict	4902	6.87	15.82	0	629	4
Local government payroll	Quarterly	90 days	4789	261.89	108.01	51	1214	242
		180 days	4789	261.89	108.01	51	1214	242
		365 days	4789	261.89	108.01	51	1214	242
		365/no subdistrict	4906	130.85	173.56	17	2283	94
Quarters in office	Quarterly	90 days	4789	8.86	6.33	1	39	7
		180 days	4789	8.86	6.33	1	39	7
		365 days	4789	8.86	6.33	1	39	7
		365/no subdistrict	4906	8.83	6.32	1	39	7
Log population	Annual	90 days	4789	10.65	0.91	9	13	11
		180 days	4789	10.65	0.91	9	13	11
		365 days	4789	10.65	0.91	9	13	11
		365/no subdistrict	4906	10.69	0.94	9	13	11

B.2 Appointment spiking: Within-district design

B.2.1 Descriptives

Figure B3 shows mean appointment counts in districts before and after head turnover. The red dashed line shows the mean appointment count for the whole appointments dataset.

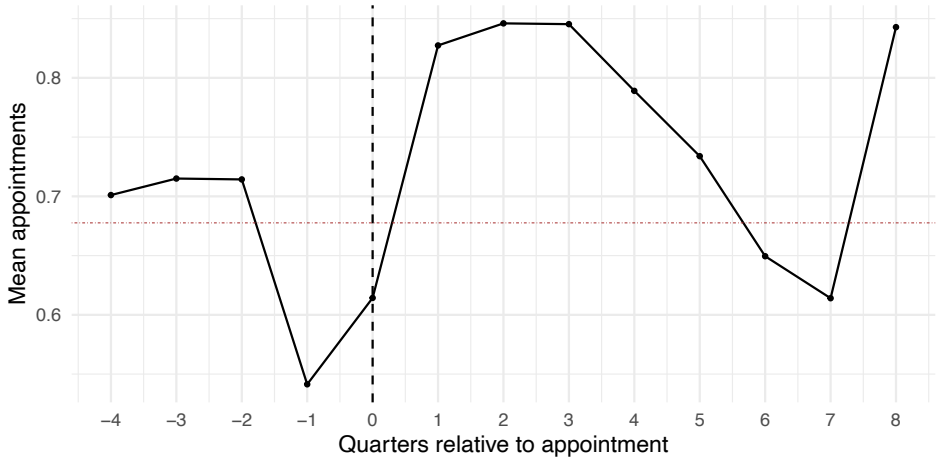


Figure B3: Mean appointment counts before and after new head appointments

B.2.2 Pre-trends

Results for the F-tests that pre-period coefficients equal zero, both excluding the quarter immediately before appointment ($H_0 : \beta_{-4} = \beta_{-3} = \beta_{-2} = 0$) and including it ($H_0 : \beta_{-4} = \beta_{-3} = \beta_{-2} = \beta_{-1} = 0$), are in Table B3.

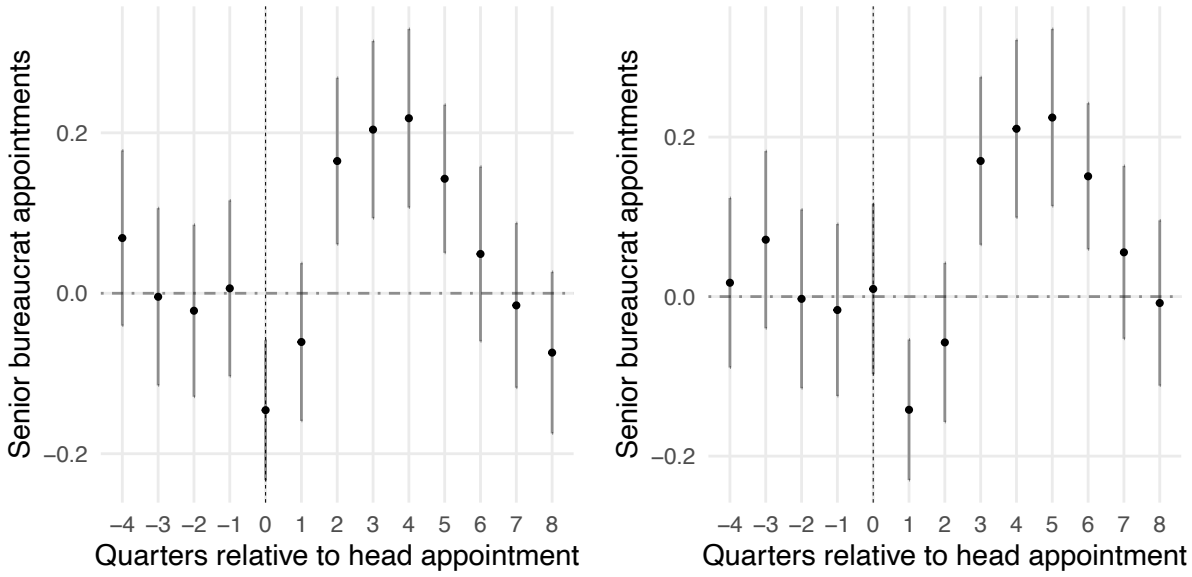
Table B3: Joint tests of pre-trend coefficients, by measurement rule

Rule	Model	Leads	F-statistic	Degrees of freedom	p-value
90 days	No controls	β_{-4} to β_{-1}	1.06	(4, 199)	0.379
		β_{-4} to β_{-2}	0.23	(3, 199)	0.877
	Full controls	β_{-4} to β_{-1}	1.12	(4, 199)	0.350
		β_{-4} to β_{-2}	0.22	(3, 199)	0.881
180 days	No controls	β_{-4} to β_{-1}	3.71	(4, 199)	0.006**
		β_{-4} to β_{-2}	0.40	(3, 199)	0.752
	Full controls	β_{-4} to β_{-1}	3.88	(4, 199)	0.005**
		β_{-4} to β_{-2}	0.40	(3, 199)	0.750
365 days	No controls	β_{-4} to β_{-1}	3.26	(4, 199)	0.013*
		β_{-4} to β_{-2}	0.15	(3, 199)	0.927
	Full controls	β_{-4} to β_{-1}	3.73	(4, 199)	0.006**
		β_{-4} to β_{-2}	0.18	(3, 199)	0.912
365/no subdistrict	No controls	β_{-4} to β_{-1}	1.83	(4, 202)	0.124
		β_{-4} to β_{-2}	0.71	(3, 202)	0.547
	Full controls	β_{-4} to β_{-1}	1.96	(4, 202)	0.101
		β_{-4} to β_{-2}	0.79	(3, 202)	0.503

Signif. codes: ***: 0.001, **: 0.01, *: 0.05, +: 0.1

B.2.3 Anticipation in the within-district design

I account for anticipation by shifting the treatment variable D_{it} back x quarters, where x is a plausible amount of anticipation. Here, I show that my conclusions hold assuming the outgoing head anticipates their departure one or two quarters before actual turnover.



(a) One quarter of anticipation

(b) Two quarters of anticipation

Figure B4: Within-district design accounting for anticipation

B.3 Appointment spiking: Within-manager design

An alternative approach instead exploits variation in the districts managers run. Sometimes, district heads are rotated to run a different district. This further allows me to hold *manager* characteristics constant and deal with concerns about selection in headship. My theory would predict that, on moving to a new district, a head appoints more than they did towards the end of their last term. I restrict the sample to 92 heads observed starting terms in at least two districts within one calendar quarter of leaving the previous one (98 transfers for 1,533 estimation rows). I estimate another multiple-event design where transfer to a different district is the treatment. The pre-period now measures the same head's appointment behaviour in the final year of their previous tenure. This model includes head as well as district and quarter fixed effects

and clusters standard errors on head.

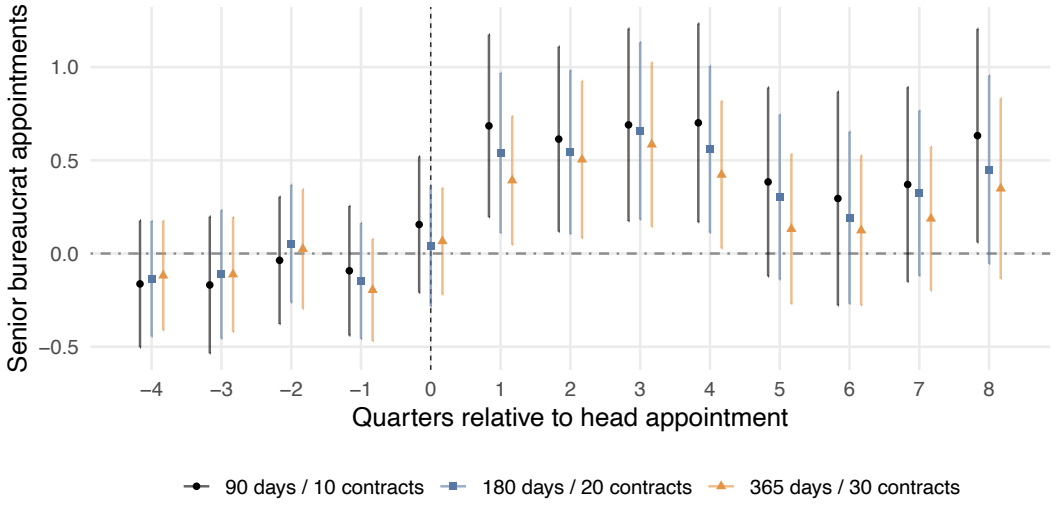


Figure B5: Event study estimates for the within-manager multiple-event model

Figure B5 shows the results. Altogether, they provide further evidence that newly appointed managers engage in excess hiring. Joint estimates are significant at the 5% level for all rules. The pre-trends are not distinguishable from 0, so there is no evidence that transferred heads appointed differently in the final year of their tenure.

B.4 Governor departure

B.4.1 Governor turnover DiD

Table B4: Joint tests of pre-trend coefficients for governor departure, by measurement rule

Rule	Model	Leads	F-statistic	Degrees of freedom	p-value
90 days	No controls	β_{-4} to β_{-1}	2.71	(4, 3665)	0.029*
		β_{-4} to β_{-2}	3.60	(3, 3665)	0.013*
	Full controls	β_{-4} to β_{-1}	3.04	(4, 3665)	0.016*
		β_{-4} to β_{-2}	4.06	(3, 3665)	0.007**
180 days	No controls	β_{-4} to β_{-1}	1.37	(4, 3665)	0.240
		β_{-4} to β_{-2}	1.34	(3, 3665)	0.261
	Full controls	β_{-4} to β_{-1}	1.46	(4, 3665)	0.213
		β_{-4} to β_{-2}	1.52	(3, 3665)	0.206
365 days	No controls	β_{-4} to β_{-1}	2.27	(4, 3665)	0.059+
		β_{-4} to β_{-2}	2.62	(3, 3665)	0.049*
	Full controls	β_{-4} to β_{-1}	2.19	(4, 3665)	0.068+
		β_{-4} to β_{-2}	2.56	(3, 3665)	0.053+
365/no subdistrict	No controls	β_{-4} to β_{-1}	1.15	(4, 3774)	0.329
		β_{-4} to β_{-2}	1.19	(3, 3774)	0.312
	Full controls	β_{-4} to β_{-1}	1.50	(4, 3774)	0.199
		β_{-4} to β_{-2}	1.50	(3, 3774)	0.213

Signif. codes: ***: 0.001, **: 0.01, *: 0.05, +: 0.1

Table B4 shows the results of joint pre-trend F-tests.

Table B5: Effect of appointing governor departure on senior bureaucrat appointments, with and without head-tenure fixed effects, by measurement rule

	Senior bureaucrat appointments					
	90 days		180 days		365 days	
	(1)	(2)	(3)	(4)	(5)	(6)
Post-departure = 1	-0.005	-0.157*	-0.024	-0.148*	-0.010	-0.091
	(0.062)	(0.073)	(0.057)	(0.065)	(0.055)	(0.058)
Controls	Full	Full	Full	Full	Full	Full
Tenure FE	Yes	No	Yes	No	Yes	No
Observations	3,676	3,676	3,676	3,676	3,676	3,676
R^2	0.000	0.004	0.000	0.004	0.000	0.002
Adjusted R^2	0.000	0.004	0.000	0.004	0.000	0.002

Clustered (reg_code) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, +: 0.1*

B.5 Alternative explanations

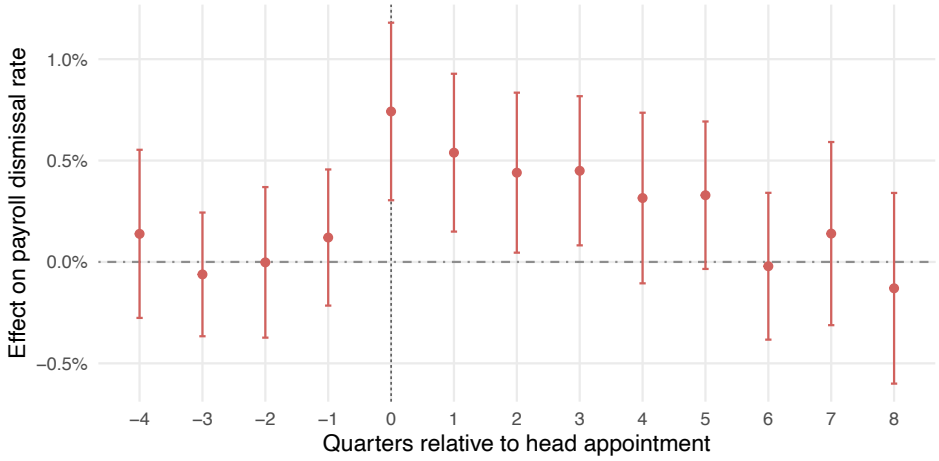


Figure B6: Event study estimates for the effect of head turnover on all-payroll separation.

Table B6: The appointment spike by election adjacency

Quarters from appointment	Election-adjacent events	Election-distant events	Adjacent − distant
4 quarters before	-0.056 (0.127)	-0.113 (0.176)	0.057 (0.198)
3 quarters before	-0.165 (0.122)	0.021 (0.147)	-0.186 (0.176)
2 quarters before	-0.213 (0.130)	-0.112 (0.135)	-0.101 (0.174)
1 quarter before	-0.379** (0.139)	-0.189 (0.151)	-0.189 (0.208)
At appointment	-0.206 (0.128)	-0.270^{+} (0.160)	0.064 (0.204)
1 quarter after	0.367*** (0.110)	0.372*** (0.110)	-0.005 (0.142)
2 quarters after	0.330** (0.107)	0.190 (0.131)	0.140 (0.160)
3 quarters after	0.325** (0.102)	0.151 (0.140)	0.174 (0.165)
4 quarters after	0.313** (0.110)	0.217^{+} (0.130)	0.097 (0.175)
5 quarters after	0.144 (0.130)	-0.058 (0.140)	0.202 (0.190)
6 quarters after	-0.178 (0.131)	-0.004 (0.130)	-0.174 (0.198)
7 quarters after	-0.022 (0.143)	0.063 (0.118)	-0.085 (0.184)
8 quarters after	0.210^{+} (0.126)	0.134 (0.124)	0.076 (0.175)
Spike (average, 1 to 4)	0.334*** (0.072)	0.232* (0.091)	0.102 (0.106)
Spike, adjacency within -2 to +3	0.311*** (0.072)	0.297*** (0.079)	0.014 (0.092)
Spike, adjacency within 4 quarters	0.312*** (0.068)	0.287* (0.145)	0.026 (0.162)
Joint pre-trend test (p)	0.332	0.756	–

*** p < 0.001; ** p < 0.01; * p < 0.05; $^{+}$ p < 0.1.

B.6 Results excluding subdistrict heads

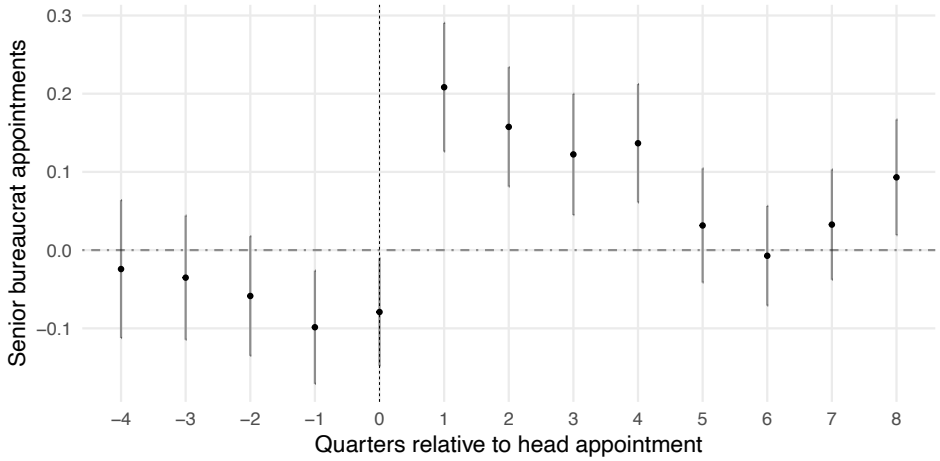


Figure B7: Event study estimates for the within-district multiple-event model (no subdistrict heads)

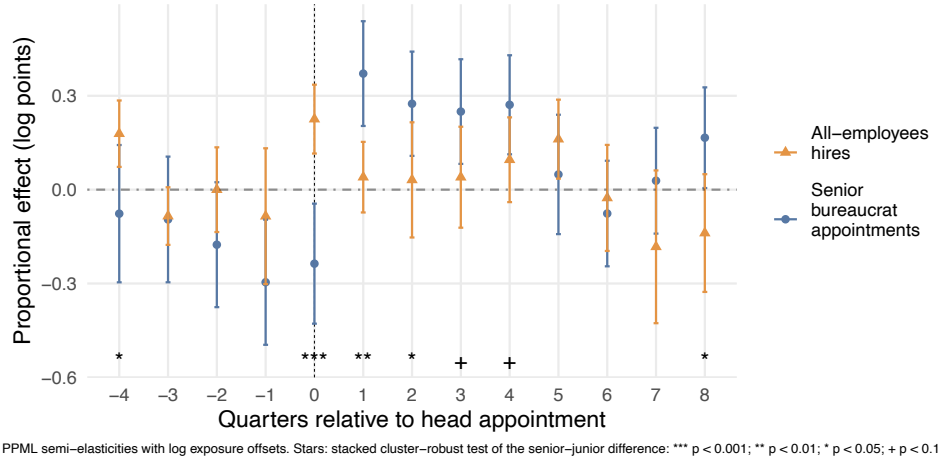


Figure B8: Proportional changes in senior bureaucrat and payroll hiring after new manager entry (no subdistrict heads; *Note*: Significance codes are two-sided p-values from a stacked regression: ***: 0.001, **: 0.01, *: 0.05, +: 0.1)

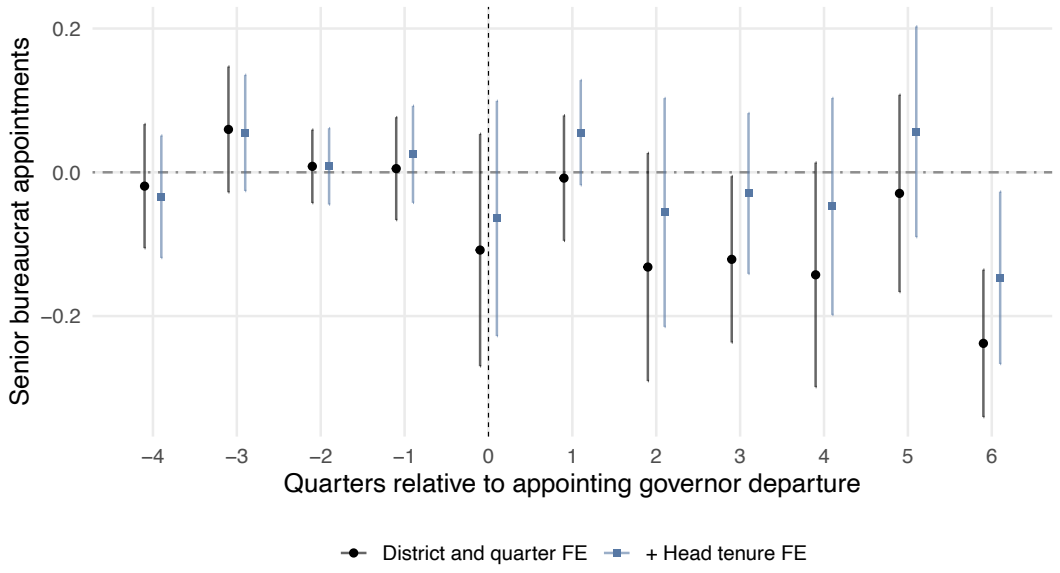


Figure B9: Event study estimates for the effect of the appointing governor's departure on district senior bureaucrat appointments, with and without controlling for head tenure (no subdistrict heads)

Supplementary material references

Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, David Altman, Fabio Angiolillo, et al. 2025. *V-Dem Codebook V15.*

Lührmann, Anna, Marcus Tannenberg, and Staffan I. Lindberg. 2018. "Regimes of the World (RoW): Opening New Avenues for the Comparative Study of Political Regimes." *Politics and Governance* 6 (1): 60–77.